

# Chapter 7

## Economies of Scale and Scope

# Learning Objectives

- 7.1 Explain the law of diminishing marginal returns.
- 7.2 Describe the importance of knowing what the cost curves look like while negotiating contracts.
- 7.3 Describe the relationship between average costs and output.
- 7.4 Determine whether returns to scale will be increasing, constant, or decreasing.
- 7.5 Identify the cost advantage of operating near a minimum efficient scale.
- 7.6 Explain the importance of looking over the life cycle of a product characterized by learning curves.
- 7.7 Describe how economies of scope between two products can be a source of competitive advantage and shape acquisition strategy.

# Chapter 7 – Main Points

- The **law of diminishing marginal returns** states that as you try to expand output, your marginal productivity (the extra output associated with extra inputs) eventually declines.
- Increasing marginal costs eventually cause increasing average costs and make it more difficult to compute break-even prices. When negotiating contracts, it is important to know what your costs curves look like; otherwise, you could end up accepting contracts with unprofitable prices.
- If average cost falls with output, then you have **increasing returns to scale**. In this case you want to focus strategy on securing sales that enable you to realize lower costs. Alternatively, if you offer suppliers big orders that allow them to realize **economies of scale**, try to share in their profit by demanding lower prices.
- If your average costs are constant with respect to output, then you have **constant returns to scale**. If average costs rise with output, you have **decreasing returns to scale** or **diseconomies of scale**.
- **Learning curves** mean that current production lowers future costs. It's important to look over the life cycle of a product when working with products characterized by learning curves.
- If the cost of producing two outputs jointly is less than the cost of producing them separately—that is,  $\text{Cost}(Q_1, Q_2) < \text{Cost}(Q_1) + \text{Cost}(Q_2)$  — then there are economies of scope between the two products. This can be an important source of competitive advantage and shape acquisition strategy.

# Warm-Up

- Have you ever shopped at Costco or Sam's Club? Why do you think they're able to offer lower prices than regular grocery stores even for the same product?

# Relating MC to MP – Review

Marginal cost and marginal product are defined as follows:

$$MC = \frac{\Delta TC}{\Delta Q} \qquad MP = \frac{\Delta Q}{\Delta L}$$

We can rewrite this as the following:

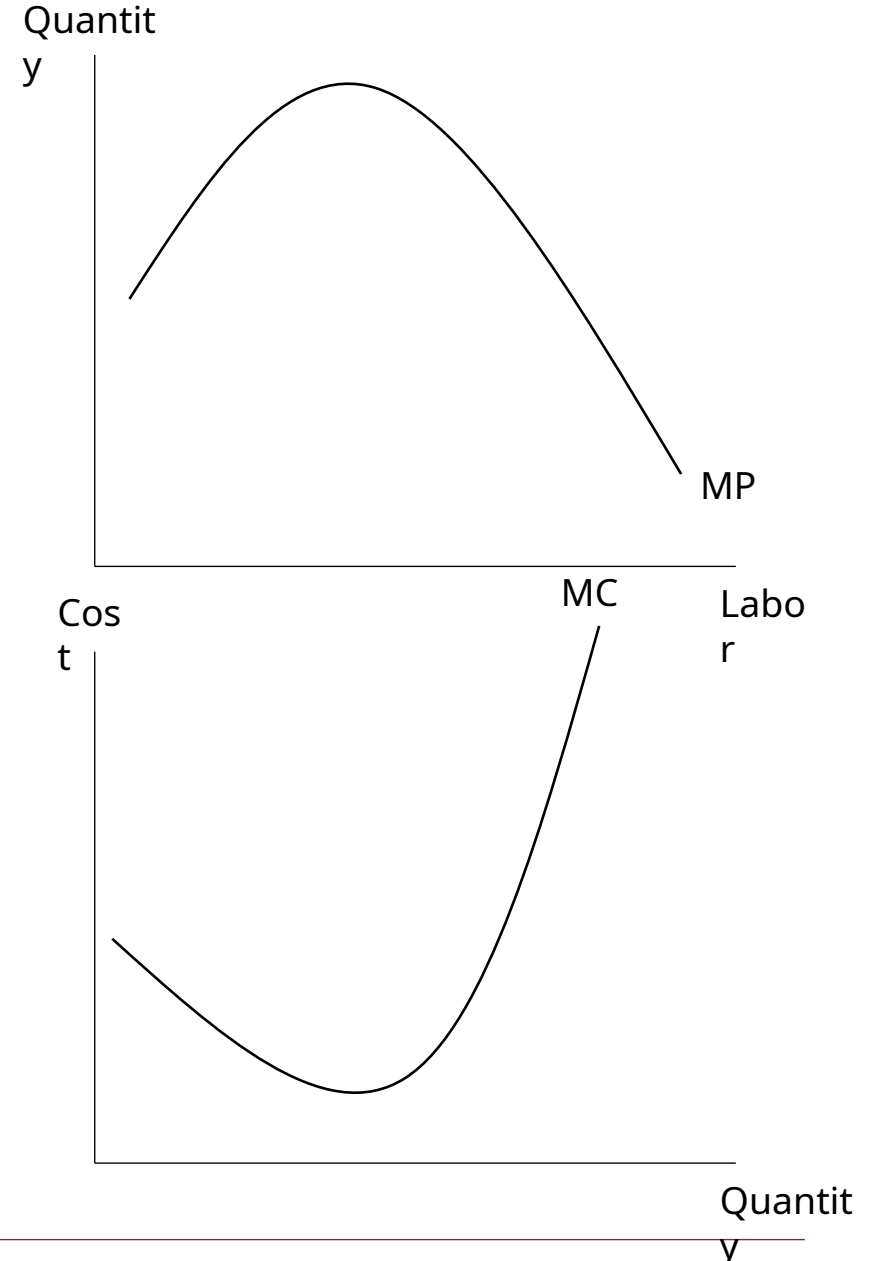
$$MC = \frac{\Delta VC}{\Delta Q} \qquad \text{Since FC do not change}$$

If we let  $VC = w \times \Delta L$  and divide  $\Delta VC$  by  $\Delta L$  we get:

$$MC = \frac{w}{\Delta TP / \Delta L} \qquad \text{So, MC is the reciprocal of MP}$$

# Relating MC to MP – Review

- As more labor is hired, the marginal product of labor increases. This causes the marginal cost of production to decrease.
- As the marginal cost decreases, it pulls down average cost.
- After diminishing marginal returns sets in, marginal costs begin to increase and begin to pull up the average cost of production.



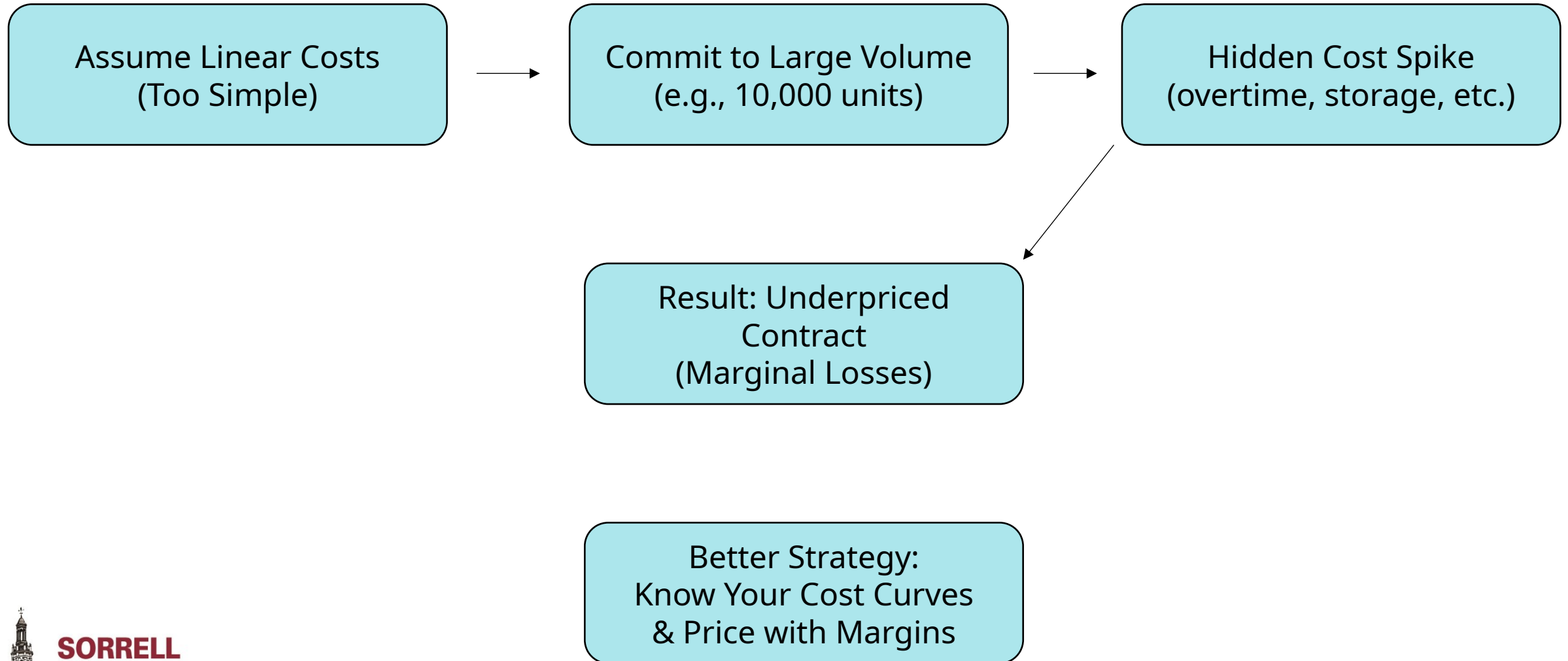
# Contracts and Cost Curves

- If you don't understand how costs behave as output scales, you risk:
  - Overcommitting to large volume at marginal losses
  - Underpricing a deal that erodes profit margin
  - Short-run costs can rise sharply beyond capacity

## Example:

- A supplier agrees to produce 10,000 units at a low fixed price, assuming linear cost increase. But after 6,000 units, overtime, storage, and logistics costs spike.
- Knowing beforehand how costs (marginal) will change allows for better contract making.

# Contracts and Cost Curves: Why Marginal Costs Matter





# Discussion Activity 1

- Imagine you're a manager negotiating a production contract. Why would relying on *average* cost lead you to underprice?
- What happens to costs once you exceed your capacity? What kind of costs suddenly spike?

# How ATC Changes as the Scale of Production Changes

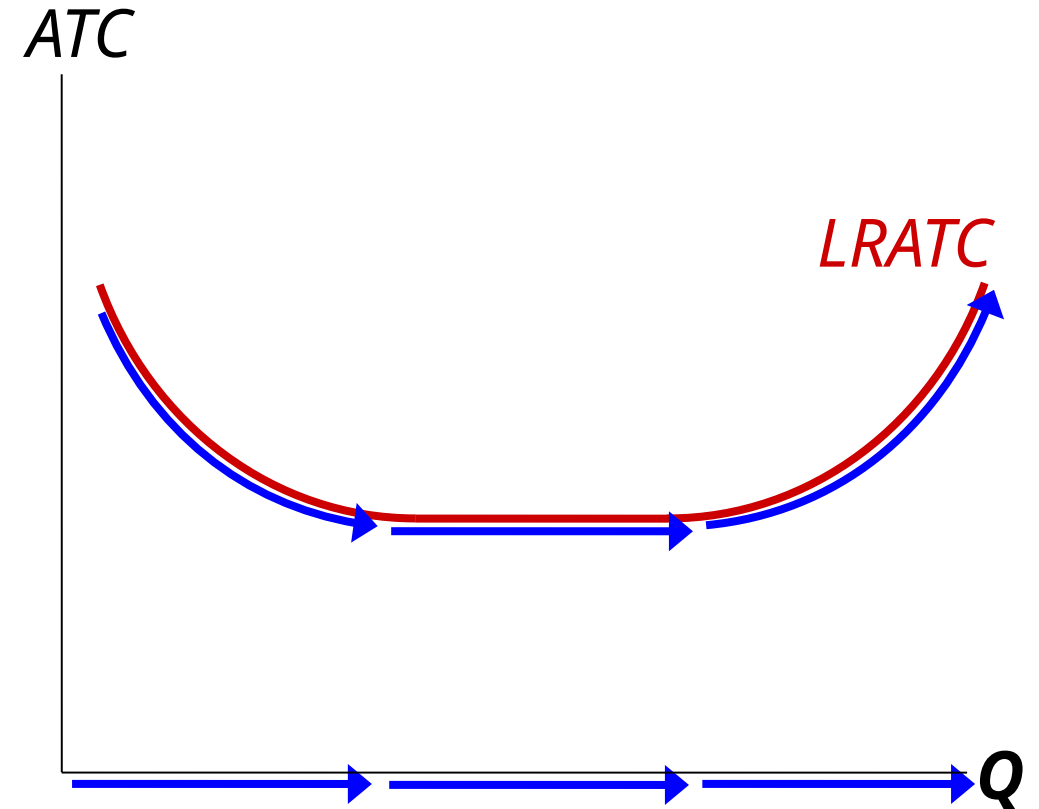
- **Economies of scale** occur when increasing production allows greater specialization: workers are more efficient when focusing on a narrow task.
  - More common when **output** is low.
- **Minimum efficient scale** is the smallest quantity of output at which a firm's average total cost is minimized in the long run
  - Fully realized economies of scale, producing more does not lower ATC, higher per-unit costs exists outside of MES
- **Diseconomies of scale** are due to coordination problems in large organizations.  
e.g., management becomes stretched, can't control costs.
  - More common when **output** is high.

# Economies of Scale

- Firms can experience economies of scale at different levels
  - Product level: specialized tools reduce cost/increase output
    - Homebrew to Craft/Micro brew
  - Plant level: Capital investment allows for higher levels of output
    - IBM invested in capital to increase its chip wafer size from 8 inches to 12 inches. This allowed a reduction in costs from \$23/wafer to \$14/wafer.
  - Firm level: Better distribution allows for lower transportation costs

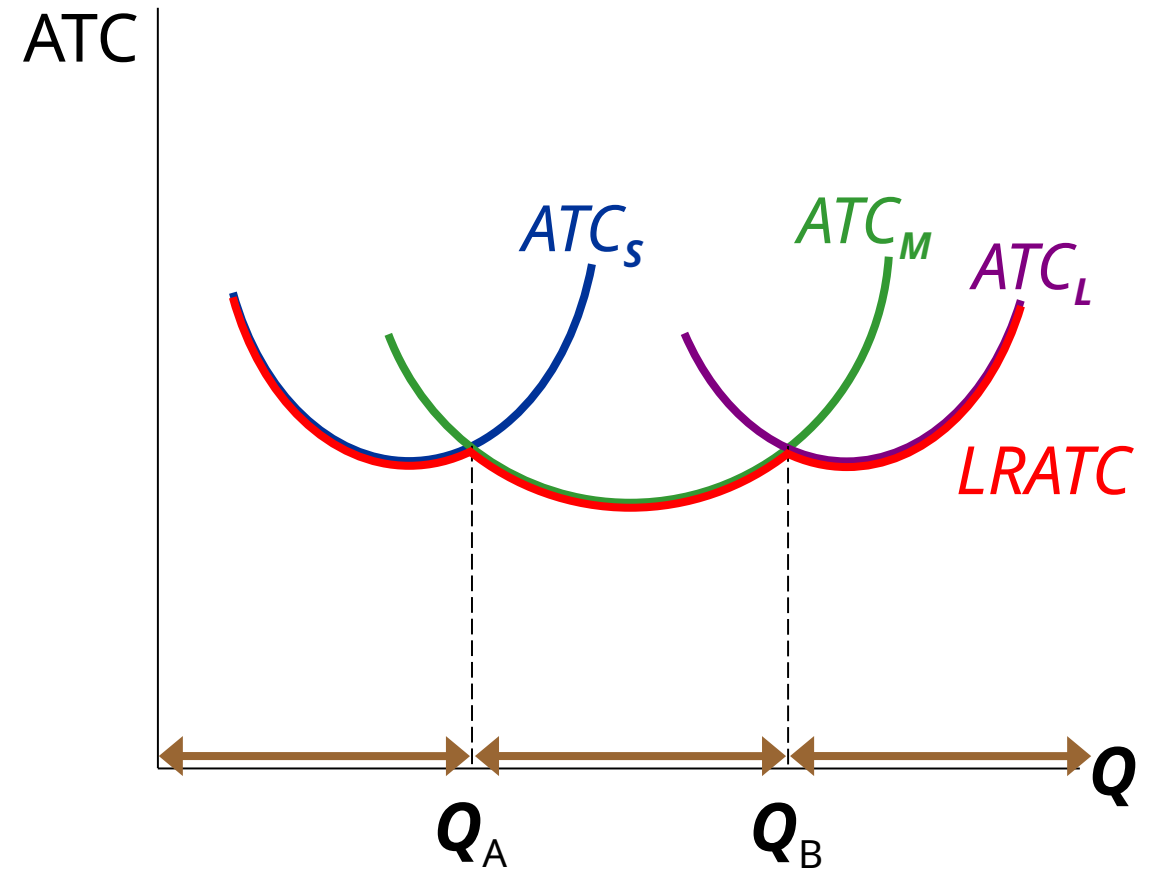
# How ATC Changes as the Scale of Production Changes

- **Economies of scale:** ATC falls as  $Q$  increases.
  - Fixed costs spread over more units
- **Constant returns to scale:** ATC stays the same as  $Q$  increases.
  - Minimum efficient scale (MES)
- **Diseconomies of scale:** ATC rises as  $Q$  increases.
  - Coordination issues, bottlenecks, etc.



# LRATC with 3 factory sizes

- To produce less than  $Q_A$ , firm will choose size **S** in the long run.
  - Below MES, underutilizing scale
- To produce between  $Q_A$  and  $Q_B$ , firm will choose size **M** in the long run.
  - At MES
- To produce more than  $Q_B$ , firm will choose size **L** in the long run.

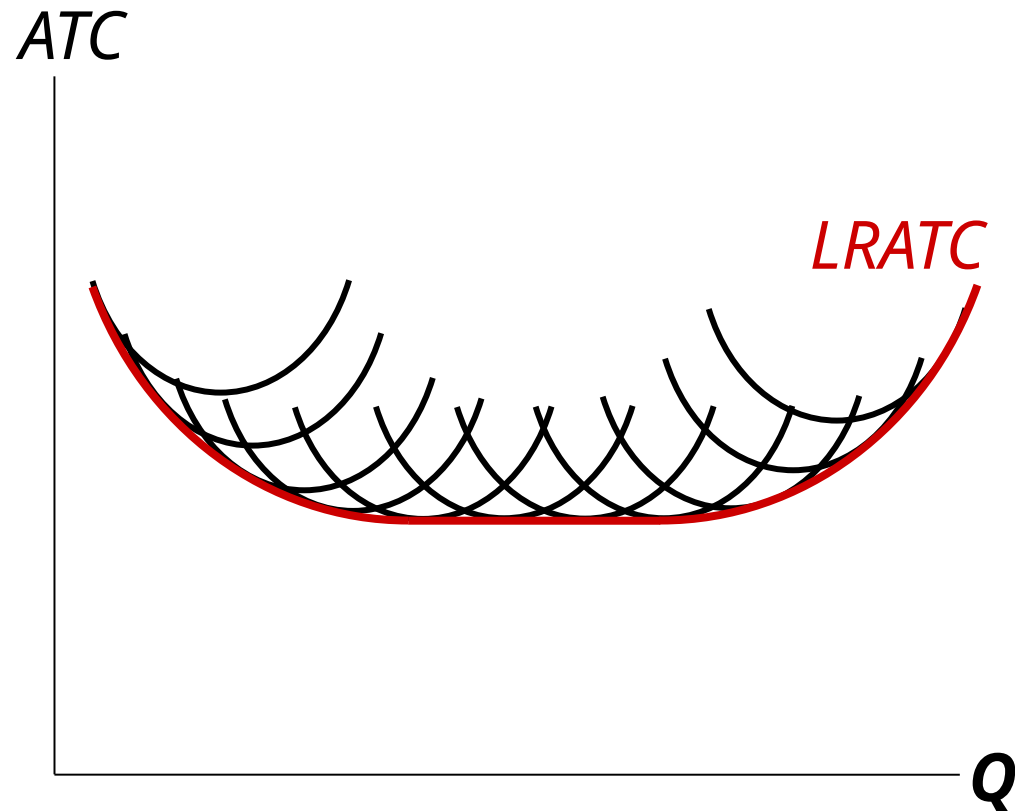


• Beyond MES, diseconomies

# A Typical LRATC Curve

In the real world, factories come in many sizes, each with its own *SRATC* curve.

So, a typical *LRATC* curve looks like this:



# Activity - Identify the Type of Returns to Scale

## Scenario:

- A factory doubles all inputs:
  - Labor: from 50 to 100
  - Capital: from 10 machines to 20 machines
- As a result, output increases:
  - Case A: from 5,000 units to 12,000
  - Case B: from 5,000 units to 10,000
  - Case C: from 5,000 units to 8,000

## Questions:

- Label each case as increasing, constant, or decreasing returns to scale.
- Why do you think they differ?

# Activity - Why Minimum Efficient Scale Matters

## Scenario:

- Two firms operate in the same industry:
  - Company A is producing at 50% of MES output
  - Company B is producing at MES
- The long-run average cost curve is U-shaped, and demand in the market is stable but limited.

## Questions:

1. Which company has the cost advantage? Why?
2. What are Company A's strategic options to reduce its average costs?
3. If market demand is too low to support MES output, what challenges arise for both companies?
4. What type of market structure might result if only one firm can operate at MES efficiently?



# Activity - Why Minimum Efficient Scale Matters

1. Which company has the cost advantage? Why?
  - Company B – because it operates at the minimum efficient scale, its average total cost is minimized, giving it pricing flexibility and higher potential profit margins.
2. What are Company A's strategic options?
  - Increase output to reach MES and lower average costs
  - Partner, merge, or outsource to share fixed costs
  - Exit the market if unable to reach cost parity
  - Focus on niche/quality strategy if cost leadership is out of reach
3. What if market demand is too low to support MES?
  - Neither firm may be able to operate efficiently
  - Price competition could lead to losses or market exit
  - May result in natural monopoly or regulated pricing
4. What type of market structure might result?
  - Natural monopoly or oligopoly
  - If only one or a few firms can reach MES due to scale or demand constraints, it limits competition and may invite regulation or barriers to entry

# Returns to Scale in the Real World

| Type of Return           | Cost Elasticity ( $E_c$ )     | Example Firm                         |
|--------------------------|-------------------------------|--------------------------------------|
| Increasing ( $E_c < 1$ ) | Costs rise slower than output | Cloud computing, Amazon Web Services |
| Constant ( $E_c = 1$ )   | Costs rise at same rate       | Chain coffee shop with stable ops    |
| Decreasing ( $E_c > 1$ ) | Costs rise faster than output | Conglomerate with bloated structure  |

# Learning Curves

- Learning Curve Concept

- A learning curve shows how production costs decline with experience. As cumulative output increases, firms become more efficient, reducing per-unit costs.
- Learning causes an inward shift in the LRAC curve due to better production knowledge.
- Learning is often mistaken for scale economies.
  - The key difference is output per period (scale economies) vs. cumulative total output

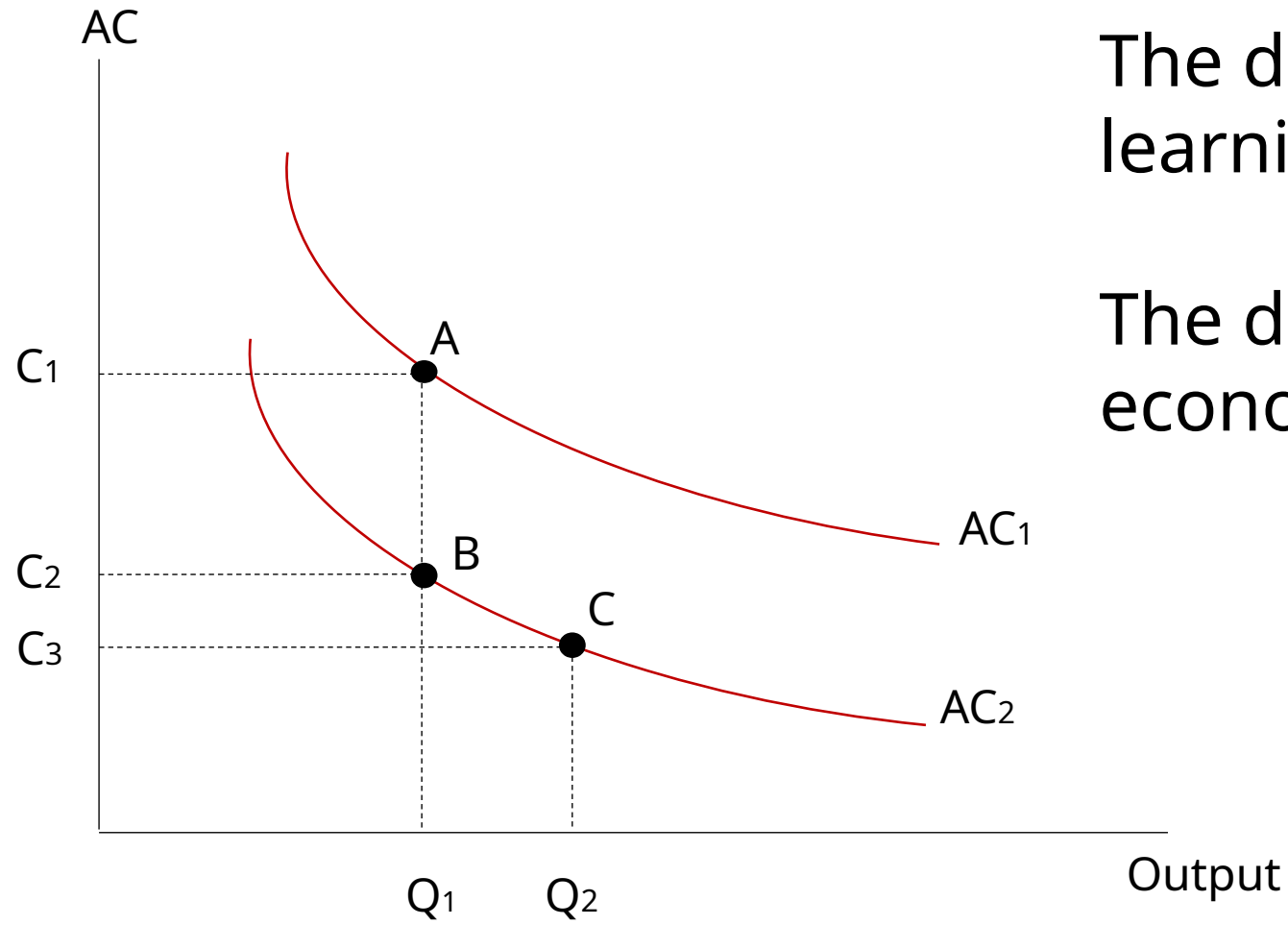
- The learning rate is found using the following:

$$LR = \left( \frac{\text{New Cumulative Average Cost}}{\text{First Average Cost}} \right) \times 100$$

- Strategic Implications of the Learning Curve Concept

- Encourages scale and early market entry
- Critical for competitive advantage in cost-driven industries
- Affects capacity planning decisions

# Learning Curve Graph



The distance AB represents the learning rate.

The distance BC represents economies of scale.

# Learning Curve Example

- Suppose a tech company assembles 10,000 smartphones at a total cost of \$1,000,000. What is their average cost per phone?

$$AC1 = TC / Q$$

$$\$1,000,000 / 10,000 = \$100$$

- As it turns out, the tech company was able to optimize their assembly line and employee training. The next 10,000 units cost only \$800,000 to produce. The learning rate is

$$AC2 = \$800,000 / 10,000 = \$80$$

$$\text{Learning Rate} = (AC2 / AC1) \times 100\%$$

$$= (80 / 100) \times 100\% = 80\%$$

The company now produces smartphones at 80% of the original cost and achieves a 20% cost reduction.

# Activity – Learning Curve Analysis

## Scenario:

- A manufacturing firm produces drones. The learning curve suggests that each time cumulative production doubles, the average cost per unit falls by 20%.

| Cumulative Output | Average Cost per Unit |
|-------------------|-----------------------|
| 100 units         | \$500                 |
| 200 units         | ?                     |
| 400 units         | ?                     |

## Questions:

- What is the average cost at 200 units?
- What is the average cost at 400 units?
- What is the total cost savings per unit between 100 and 400 units?
- What are the strategic implications of learning curves for:
  - Pricing strategy?
  - Early losses?
  - Barriers to entry?

# Activity – Learning Curve Analysis

1. At 200 units:

- $\$500 \times (1 - 0.20) = \$400$

2. At 400 units:

- $\$400 \times (1 - 0.20) = \$320$

3. Cost savings per unit from 100 → 400:

- $\$500 - \$320 = \$180$  saved per unit

4. Strategic implications:

- Firms may price low early to drive volume and lower future costs.
- Early entrants gain cost advantages competitors may not match.
- Learning curves can act as barriers to entry.

# Discussion Activity 2

- Can you think of a product where the more it's produced, the cheaper or better it becomes? Why does that happen?
- Consider software, phones, food production. Is it due to practice, scale, or both?



# Learning Curve v. Economies of Scale

| Feature               | Learning Curve                    | Economies of Scale                 |
|-----------------------|-----------------------------------|------------------------------------|
| Cost reduction source | Worker experience & process gains | Higher output spreading fixed cost |
| Time-based?           | Yes (more cumulative output)      | No (based on output level)         |
| Applies to?           | New products, tech firms          | Most manufacturing, logistics      |

Both reduce cost, but for different reasons. Use learning curves to plan launches, economies of scale to optimize production.

# Economies of Scope

- Economies of Scope Concept
  - Scope economies are cost advantages that stem from producing multiple outputs.
    - For instance, an airline will transport passengers and cargo
    - If firms are taking advantage of economies of scope, then they have realized a cost advantage such that producing two products together provides a cost reduction over producing them separately.
  - Big scope economies explain the popularity of multi-product firms.
  - Without scope economies, firms specialize in only one product
- Exploiting Scope Economies
  - Scope economics often shape competitive strategy for new products.
  - A craft brewer may notice that a slight change in a popular recipe yields a slightly different beer that is equally profitable and helps increase market share

# Finding Economies of Scope

$$S = \frac{C(Q1) + C(Q2) - C(Q1 + Q2)}{C(Q1 + Q2)}$$

$S$  = degree of economies of scope

$C(Q1)$  = cost of producing first product alone

$C(Q2)$  = cost of producing second product alone.

- Suppose that  $S = 0.25$ . 25% less expensive to produce 2 goods together than separately.

# Activity – Measuring Economies of Scope

Scenario:

- A company produces Product A and Product B. The cost data is as follows:
    - Cost of producing only Product A = \$400,000
    - Cost of producing only Product B = \$300,000
    - Cost of producing both A and B together = \$650,000
1. What is the total cost of producing A and B separately?
  2. What is the cost saving from producing both together?
  3. Calculate the Economies of Scope (S):
  4. Interpret the result – Is there an economy of scope?

# Activity – Measuring Economies of Scope

1.  $\$400,000 + \$300,000 = \$700,000$
2.  $\$700,000 - \$650,000 = \$50,000$  saved
3.  $((400,000 + 300,000) - (650,000)) / (650,000) = 7.69\%$
4. Yes. While not large, a positive result ( $S > 0$ ) means there are economies of scope, suggesting cost advantages from joint production. This could inform acquisition, bundling, or integration strategy.

# Discussion Activity 3

- Why do companies like Amazon, Google, or Apple sell so many different products and services? Is it always about profit?
- Think about shared infrastructure, customer relationships, or bundling. Is there a cost or reputation risk to diversification?

# Case Study: Wells Fargo and Cross-Selling

- [Wells Fargo takes advantage](#) of providing many banking products to the same consumer. This is called cross-selling.
- It is also economies of scope, as Wells Fargo finds it cheaper to provide multiple products instead of one.
- Though, with its cross-selling practices, Wells Fargo has a problem.
- They faced \$185 million in fines for opening [565,000 credit card](#) accounts and [2.1 million accounts total](#) that customers may not have wanted.
- The problem is easily identified. Wells Fargo created accounts that customers didn't want.
- But think about it from the consumer's viewpoint. Are they really acting in their own self-interest by getting all their banking/financial products from one company?

# Key Takeaways

- Law of diminishing marginal returns: Adding more input eventually lowers additional output (productivity), increasing marginal cost.
- Economies of scale occur when average total cost (ATC) decreases as output increases.  
→ Fixed costs are spread over more units.
- Diseconomies of scale: ATC increases at high output levels due to coordination problems and inefficiencies.
- Returns to scale explain how output responds to input changes:
  - Increasing → Output grows faster than inputs
  - Constant → Output grows proportionally
  - Decreasing → Output grows slower than inputs
- Minimum Efficient Scale (MES): The smallest output level where ATC is minimized.  
→ Operating near MES creates a competitive advantage.
- Learning curves: Costs decrease over time as firms gain experience, especially relevant in tech and manufacturing.
- Economies of scope: Cheaper to produce multiple products together than separately.  
→ Firms benefit from shared resources (marketing, distribution, R&D).
- Real-world strategy: Firms use scale and scope advantages to reduce cost, shape pricing, and guide M&A decisions.



# Knowledge Check

1. What causes average costs to rise at very high levels of output?
2. What's the danger of negotiating a large contract without understanding your cost curves?
3. How does a learning curve differ from economies of scale?
4. What is MES (minimum efficient scale)?
5. Give one example of economies of scope in the real world.

# Knowledge Check Answers

1. Diseconomies of scale—coordination problems, inefficiencies
2. You might underprice and lose money once marginal costs rise
3. Learning curves involve improvement over time; economies of scale relate to current output
4. MES is the output level where ATC is minimized
5. Amazon using same warehouse for books, toys, electronics; Google bundling services

# Consultant's Challenge: Scaling Smart

## Scenario:

- You're a consultant helping a startup that manufactures custom home fitness equipment. Business is booming, but their costs are rising faster than revenue and they don't know why.

## Your Task:

1. Where can they reduce costs as they grow?
2. Where might they hit diminishing returns and how should they respond?